SciFigBench

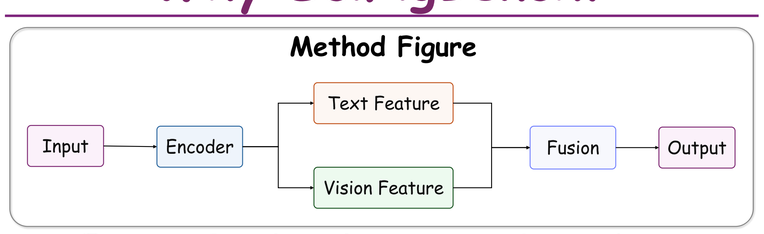
A Benchmark for Scientific Method Figure Understanding

Lijia Wu[1,2], Zheng Li[3]✉, Yuxuan Liu[1,2], Jingwen Zhang[1,2], Penghai Zhao[1,2], Xiang Li[2,1]✉

[1]VCIP, CS & AAIS, Nankai University [2]NKIARI, Shenzhen Futian [3]Li Auto.

EMNLP2026 BUDAPEST

Why SciFigBench?



Existing Benchmarks: Coarse-Grained QA

Question : How many features contribute to the final output?

: One feature. ⊗

Why did it only answer one feature?

SciFigBench (Ours): Diagnostic QA

Question : Which feature modules are direct inputs to the Fusion module?

: Vision Feature. ⊗

GT : Text Feature + Vision Feature. ✓

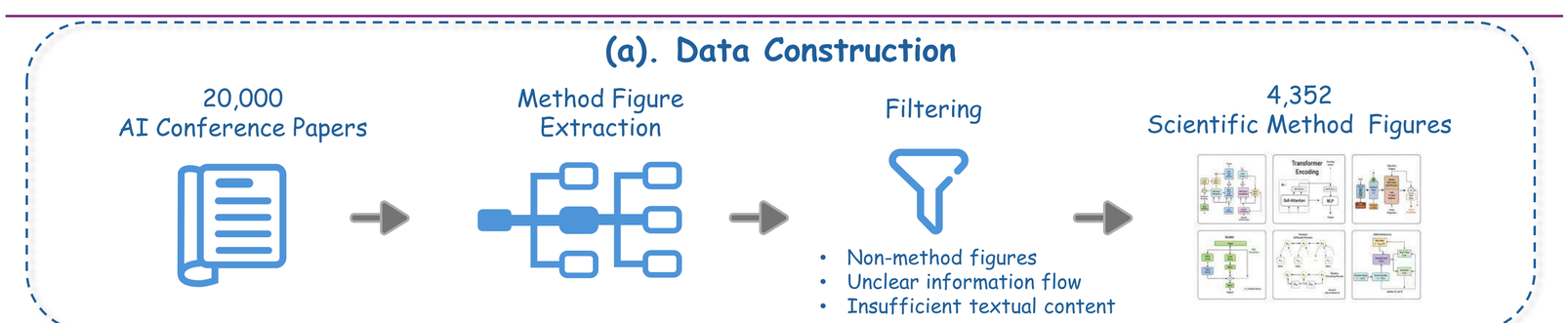
It missed the 'Text Feature' input!

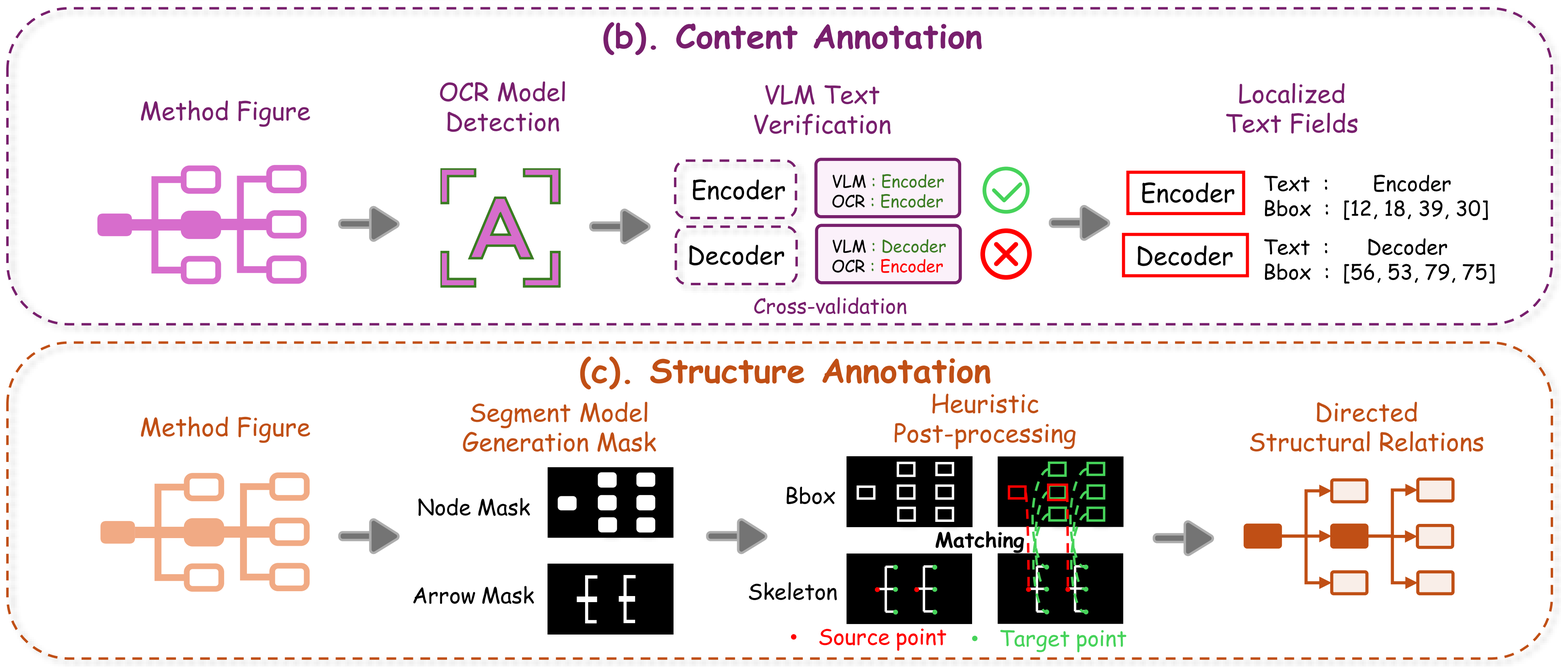
SciFigBench probes three core abilities:

- Can your VLM read text across modules?
- Can it understand arrow connections?
- Can it trace information flow?

Discover Your VLM's Limits!

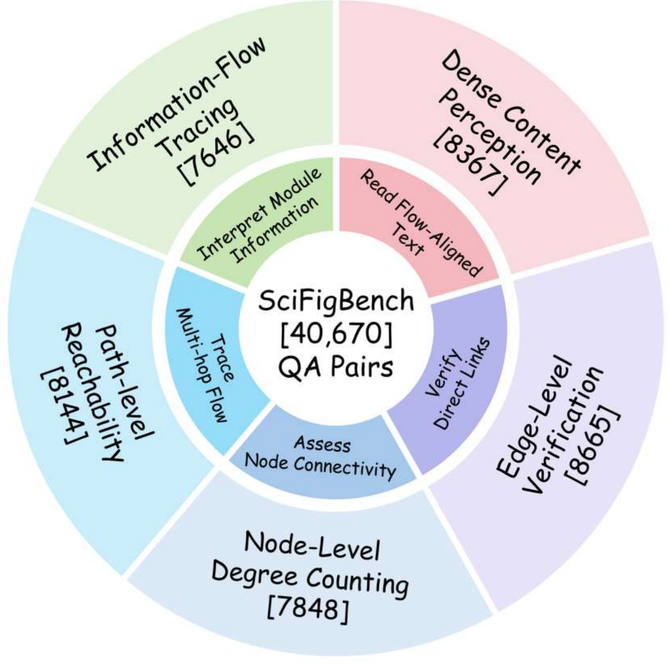
Benchmark Construction

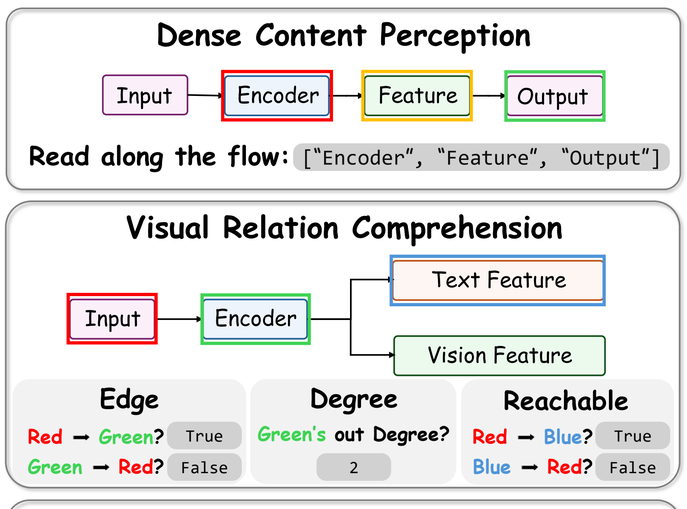


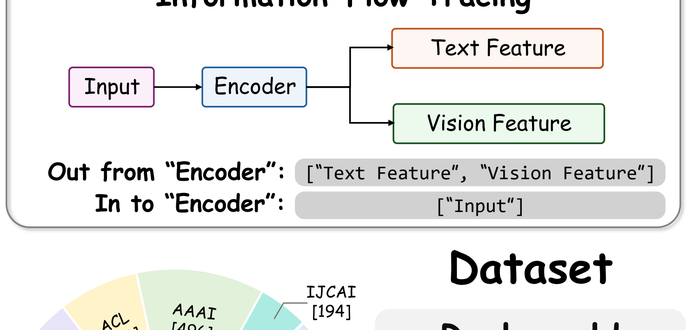


Automated Pipeline | **Human Verified** | **Public Release**

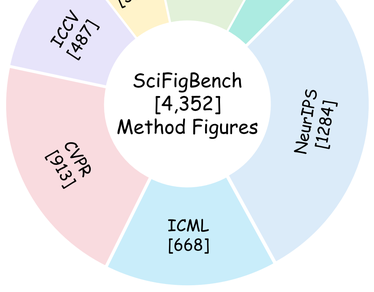
SciFigBench at a Glance







Dataset

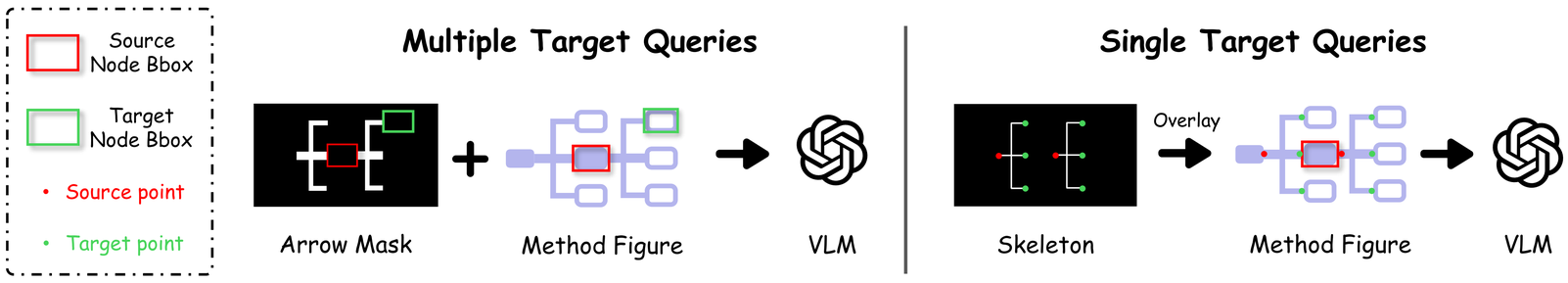


Real-world Method Figures

7 AI Conferences

2024-2025 Recent Papers

SciFigCue: Making Arrow Cues Explicit



VLMs often **miss** fine-grained arrow cues in raw method figures.
SciFigCue marks arrow locations and directions —— training-free.

Experiment Results

Model	Size	DCP	Edge	Degree	Reach.	IFT
Random choice	–	–	50.0	–	50.0	–
Closed-source models						
Gemini-2.5-Flash	–	72.7	66.0	42.3	55.3	42.4
Gemini-3.1-Flash-Lite	–	71.2	67.3	**55.8**	54.1	**46.1**
GPT-5.5	–	76.8	55.8	34.7	51.8	38.6
Open-source models						
InternVL3.5-8B	8B	45.9	44.7	35.8	41.5	27.2
InternVL3.5-38B	38B	52.2	47.8	40.9	45.4	26.4
LLaVA-OV-1.5-8B	8B	54.8	45.4	34.5	40.2	24.5
Qwen3.5-VL-2B	2B	19.8	46.2	22.0	49.2	16.3
Qwen3.5-VL-9B	9B	61.2	50.8	41.4	46.3	32.8
Qwen3.5-VL-27B	27B	70.9	59.2	50.9	50.2	43.6
SciFigCue (ours)						
SciFigCue (Qwen3.5-VL-9B)	9B	73.0 (+11.8)	56.6 (+5.8)	44.1 (+2.7)	52.0 (+5.7)	36.0 (+3.2)
SciFigCue (Qwen3.5-VL-27B)	27B	**82.4** (+11.5)	**70.1** (+10.9)	51.1 (+0.2)	**56.0** (+5.8)	42.1 (-1.5)

Main results on SciFigBench. All values are percentages; higher is better.

Scaling Isn't Enough

+51.1 DCP | +1.0 Reach.

Qwen3.5-VL 2B->27B (Δ pts)

Path Reasoning Near Chance

55.3%

Best baseline Reachability accuracy

Random choice: 50.0%

Model	K=1	K≥2 Avg.	Gap
Qwen3.5-VL-9B	76.0	45.5	30.5
+ SciFigCue	84.0	61.5	22.5
Qwen3.5-VL-27B	75.9	65.8	10.1
+ SciFigCue	85.2	79.7	5.5

K-wise Dense Content Perception Analysis.
Multi-region reading is harder; **SciFigCue** narrows the gap.

Benchmark Your VLM!

Dataset | Get Started! | WeChat